

DEEPAK SAI ARE

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PROFESSIONAL EXPERIENCE

Gen AI Engineer, Elsevier 01/2026 – Present | USA

- Architected a **multi-agent** document-understanding system using **LangGraph**, **CrewAI**, and **GPT-4o**, integrating a **hybrid-search RAG** pipeline with **LlamalIndex** and **Pinecone** to accurately process 6,000+ internal documents.
- Composed **Model Context Protocol (MCP)** servers to standardize **REST API** and database tool access for **AI agents**, and **fine-tuned** open-source models using **LoRA** to increase document classification accuracy to 89%.
- Deployed solutions on **AWS ECS Fargate** via **FastAPI**, establishing **RAGAS** evaluation and **Langfuse** observability to trace **latency** bottlenecks and reduce **inference costs** by 15-20%.

AI/ML Engineer Intern, Optum 05/2025 – 08/2025 | USA

- Prototyped foundational **generative AI workflows** by building **Python data pipelines** to extract, clean, and ingest proprietary healthcare records into **vector databases** for **semantic retrieval**.
- Integrated foundational **LLMs** with custom **REST APIs** to develop specialized planning **agents**, validating early-stage architecture for document summarization and insights.
- Containerized machine learning** services with **Docker** and processed early-stage solutions to **AWS ECS** enabling evaluation of **inference latency**, context precision, and resource allocation.

Machine Learning Engineer, HDFC 05/2022 – 12/2023

- Designed an ensemble **anomaly detection** system combining **XGBoost** and **neural networks** with **Scikit-learn** preprocessing, improving accuracy by 18% and cutting false positives by 12%, while also **fine-tuning BERT** models for complex **NLP** text classification and **entity extraction**.
- Engineered scalable **PySpark** and **SQL data pipelines** to process 1M+ daily transactions, integrating **MLflow** for experiment tracking, **model versioning**, and **ETL validation** to prevent silent **model drift**.
- Established automated **CI/CD** pipelines to deploy **Docker-based machine learning models** on **AWS ECS**, exposing capabilities through independent **microservices** and using partitioning strategies to maintain strict production **latency** SLAs.

SKILLS

Languages Python, Java, SQL, Swift, JavaScript, HTML	Backend Frameworks & Technologies Spring Boot, Django, FastAPI, REST APIs, Jakarta EE, Spring Data JPA, Apache Tomcat, AJAX, CORS
Databases PostgreSQL, MySQL, Oracle, Pinecone (Vector DB), PySpark	Cloud, DevOps & MLOps AWS (ECS, EC2, VPC), Docker, CI/CD, Git, MLflow, Langfuse, RAGAS
Architecture & Principles Microservices, DAO Pattern, Least-Privilege, Scalability, Referential Integrity, CRUD Operations	AI & Machine Learning Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangGraph, XGBoost, Scikit-learn, BERT, LoRA, Multi-Agent Systems

ACHIEVEMENTS

Medscor AI - Full Stack and AI system 10/2025

- Pioneered an AI-powered evaluation system using **AWS Bedrock (LLMs)** to convert instructor rubrics into structured **reasoning-chain** criteria, enabling self-regulated assessment of medical student oral case presentations with explainable feedback generation.
- Implemented rubric-to-evidence mapping linking scores to transcript spans and **clinical knowledge graph** nodes, ensuring **hallucination-free outputs**, traceability, and fairness in **LLM evaluation**.
- Collaborated in a 9-member team to build an **end-to-end AI evaluation pipeline**, winning the **AWS Track** among 20+ teams at Rutgers Health Hackathon 2025.

PROJECTS

Predicting Gamers Churn - Data Engineering and Backend Automation 12/2024

- Created and evaluated **machine learning models (Logistic Regression, Random Forest, XGBoost, Deep Neural Networks, LSTM)** for churn prediction, achieving 0.47 **F1-score** (precision-recall optimization) on imbalanced data, outperforming a baseline model.
- Built an end-to-end **data pipeline** using **Python**, **REST APIs**, and graph-based (BFS) traversal, processing 10,000+ records with rate limiting and checkpointing for scalable data ingestion.
- Engineered **time-series features** and behavioral features with strict temporal splits to prevent data leakage, and initiated **Streamlit** dashboard for real-time inference, model comparison, and **predictive analytics**.

Detecting Backdoor Attacks - Multi-layer Activation Clustering 11/2024

- Built a backdoor-detection workflow on CIFAR-10: crafted trigger attacks, **fine-tuned** contaminated **ResNet-18** models, and processed ~50K clean plus 1.5K poisoned images end-to-end.
- Streamlined multi-layer **feature extraction** by hooking four network blocks, exporting .npy batches, and running 2-D **PCA + K-Means** with a 0.40 silhouette alert threshold to surface anomalous activation clusters.
- Delivered reusable defense tooling by pairing Spectral Signature with **Activation Clustering**, logging attack-success metrics and generating artifacts for rapid hardening; results show near-disjoint clean/backdoor clusters from early to late layers.

EDUCATION

Master of Science in Computer Science 01/2024 – 01/2026 | New Brunswick, NJ, USA
Rutgers University - GPA : 3.7/4.0